

Insertion Sort

Definition

Insertion Sort is a comparison-based sorting algorithm that builds the sorted array **one element at a time**. Each new element is compared with the elements before it and inserted into its correct position.

Algorithm

1. Start from the **second element**.
2. Store the current element as key.
3. Compare key with the elements before it.
4. Shift elements greater than key one position to the right.
5. Insert key in its correct position.
6. Repeat until all elements are sorted.

OUTPUT:

```
Enter the number of elements: 5
Enter the elements:
5
7
9
2
6

Sorted Array:
[2, 5, 6, 7, 9]
Execution Time: 1.9311904907226562e-05 seconds

Time Complexity:
Best Case   : O(n)
Average Case: O(n^2)
Worst Case  : O(n^2)
Space Complexity: O(1)

...Program finished with exit code 0
Press ENTER to exit console.█
```

Time Complexity

- **Best Case:** $O(n)$ — array is already sorted
- **Average Case:** $O(n^2)$
- **Worst Case:** $O(n^2)$ — array is in reverse order

Space Complexity

- **Auxiliary Space:** $O(1)$
- It is an **in-place sorting algorithm**.